Sweatshop on P-Chess: the learning course

First unit (rook endgames): research, course screens, the Fathom tablebase helper and its cleanups; GPT-6.1 Terra lineup, lane `learning`

Foreman, session `sweatshop/learning/2026-11-07-2234`

8 November 2026

Your turn

Needs you:

1. **PR #44 has red CI.** Job `tbprobe-windows` has failed on every push since LEARN-010 merged: the Fathom helper built on Windows prints nothing from the accented, spaced folder. The session review (below) took it as a finding; see the chat summary for its outcome. `windows-export` also failed on the last two pushes, on a 504 from the Godot download.
2. **Confirm or reverse the probe design.** RES-001 (`Needs your call`) showed that Stockfish's UCI never gives each move's distance to zeroing. Its review reserved the change of spec L5 for you, but the proxy split LEARN-006 and created LEARN-010, a native helper built from Fathom (MIT), with L5 rewritten in `703fe57`. Everything after it depends on it.
3. **Review: human** on LEARN-004 (stage name "Endgame foundations", the 36 positions, line lengths, difficulty), LEARN-009 (Home's look) and CLEAN-020 (Home details).
4. Merge PR #44 when the above are settled.

Nothing for you: 16 tickets at `Approve` with `Review: agent`: CORE-007, CLOCK-003, CI-005, LEARN-001, 002, 003, 005, 006, 007, 008, 010, REV-011, CLEAN-016, 017, 018, 019.

Hand tests (the lane's feature is finished; every ticket is `done`):

- ☐ Fresh profile, open Course: the rating dialog has no skip (Escape, click outside); 1600 chess.com; the map shows the unit as new; Profile shows the same sentence.
- ☐ Lucena: a correct 3-move line shows √ on all three, unit page 1/12 and 100% first try; a slow detour stops at that move, and a passing retry counts as solved, not first try.
- ☐ A losing move in a hold play-out: the end card at once, with that move and a holding move (needs a real Maia: this Linux box has no Lc0).
- ☐ A real game that reached rook and pawn against rook and lost a draw: the marker in Review, its link to the unit, "From your game" and Continue.
- ☐ Windows: the tablebase probe from RES-001's procedure, the release unzipped under an accented folder, and the app restart with the saved clock choice (LEARN-002).

What changed

PR #44, 20 tickets (14 planned, 6 opened during the run).

RES-001: research only. The 40 Syzygy files the unit needs (45.68 MiB), the Stockfish 19 probe, Maia 1500/1900 on 20 sampled positions, and the finding that UCI gives no per-move distance to zeroing. `docs/research/tablebase.md`. **CORE-007**: a game keeps the position it started from, and Maia's requests use it, so play-outs can start mid-game (`core/game.gd`, `bot/maia_player.gd`). **CLOCK-003**: a 5+5 clock and a "none" clock that shows blank and never flags (`core/clock.gd`). **CI-005**: `scripts/fetch-engine.ps1` fetches and checks the 40 tables into `engine/tablebase/`, and the release requires them. **LEARN-001**: entering the course asks for your rating once and turns it into the Maia level; Profile edits it (`ui/rating_dialog.gd`, `core/course.gd`). **LEARN-002**: the course map and the unit page, with progress and a saved clock choice. Reopened once: a

pending review check hid a failed one, so "needs review" vanished; the cause was the status selection in `core/course.gd`. **LEARN-003**: the theory screen, with the goal, search terms to copy and the path to Solving (`ui/theory.gd`). **LEARN-004**: the unit's content, 12 practice and 18 test positions plus 6 play-outs, each checked against the tablebase (`data/course/rook-endgames.json`). **LEARN-005**: the Solving board, where you enter a line with legal moves without the pieces moving. Reopened once: move numbers were wrong when Black starts, and Enter wiped an unfinished line while Clear had focus, both in `ui/solving.gd`. **LEARN-010**: a small native helper built from Fathom answers each move's result and distance to zeroing over the same pipe as the engines (`native/tbprobe/main.c`, `engine/tablebase.gd`). Reopened twice: the adapter let malformed or out-of-order helper lines through, during a search and then during start-up, because `engine/uci.gd` drops lines it is not waiting for. **Risk:** its Windows build fails in CI. **LEARN-006**: grading a line against the tablebase and the feedback after it (`core/grader.gd`). **LEARN-007**: play-outs against Maia from the course position, each move judged before Maia replies; they never enter History or stats (`ui/playout.gd`). **LEARN-008**: the unit's pass test, against a perfect tablebase opponent, with retries and mastery (`ui/pass_test.gd`). **LEARN-009**: Home's Continue with its position preview, and the 7- and 30-day review checks (`ui/home.gd`). **REV-011**: Review marks a move that threw away a rook-and-pawn-against-rook result and sends that position to the unit as "From your game" (`ui/review.gd`). **CLEAN-016 to 020**: the rating dialog reuses the shared validator; course saves share one write sequence; the hold rule text follows its setting; pass-test play-outs share one teardown; Home's icons, Continue arrow and course summary match the design.

Record

$39.53 at API list price over 50 stages, all priced: `gpt-6.1-terra xhigh` implementing $15.55 (21), `gpt-6.1-sol xhigh` implementing $6.94 (5) and reviewing $17.04 (24). Codex quota: 10 points of the weekly window from a 37% baseline (22:03), no reset: Terra 4, Sol 6 (1 implementing, 5 reviewing); three threads outside the stages spent 0. The total is exact; each stage is ±1 point. The proxy (`opus-5.5 xhigh`, Claude) is not in these numbers.

What went wrong

The Windows side of the new tablebase helper was never run before approval. LEARN-010 was split out mid-run by the proxy, built and tested on Linux only, and approved with Windows "pending". The windows-latest smoke job it added then failed on every push, and nobody read CI until the run ended.

1. Make a red check on the session PR stop the driver, or at least show in its line: the job ran 7 times while 9 more tickets merged on top.
2. When a ticket adds a CI job for a platform the machine lacks, its review should wait for that job on a pushed branch, not approve on "pending".
3. Fix the driver's test-file pattern `[._-](test|spec)\.[^/.]+$`: it calls `ui/pass_test.gd` a test and parked a valid LEARN-008 handoff (unparked by hand).

Registro

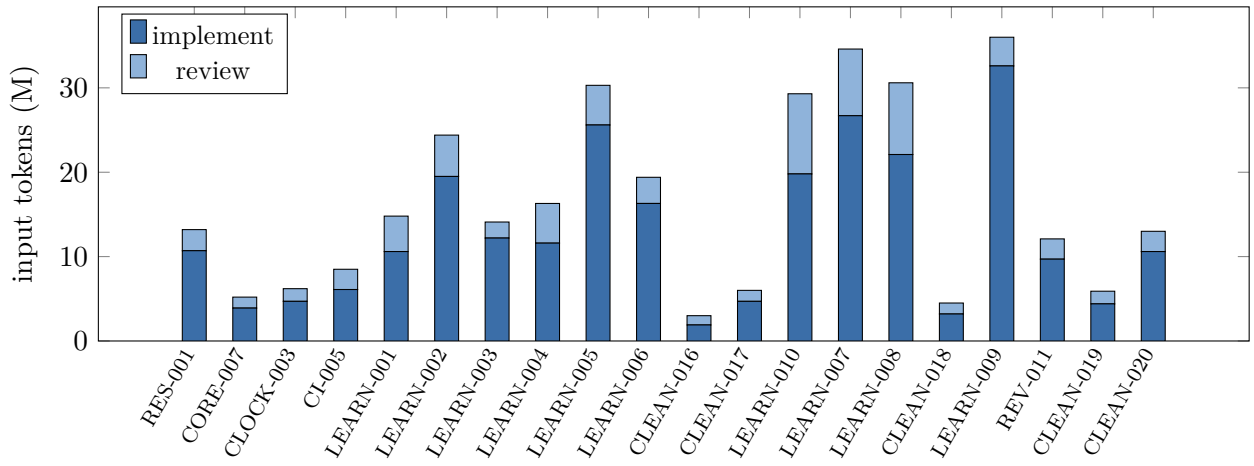
Timeline

7 Nov 22:37	Clone made from origin on the Linux box; driver launched (default lineup). Reuses the session branch a killed Windows driver had opened (empty commit `6b92979`).
22:38–00:00	RES-001, CORE-007, CLOCK-003, CI-005, LEARN-001 merged, none reopened.
00:01–00:36	LEARN-002 reopened once (status precedence), merged.
00:36–02:06	LEARN-003, LEARN-004 (review adds a missing assertion), LEARN-005 (reopened once).
02:06–02:21	LEARN-006 on the hard model asks; proxy splits off LEARN-010; no code.
02:21–02:43	CLEAN-016, CLEAN-017 merged.
02:43–03:26	LEARN-010 reopened twice; driver parks it and ends (PR #44, 11 tickets).
03:28	Foreman's reopen diagnosis and consolidated review committed; driver relaunched.
03:29–04:17	LEARN-010 merged on round 3; LEARN-006 merged.
04:17–05:34	LEARN-007 merged; LEARN-008 parked by the test-file false positive; CLEAN-018 merged; driver ends.
05:35	LEARN-008 restored to `to-review`; driver relaunched.
05:35–07:24	LEARN-008, LEARN-009 merged; REV-011 asks, proxy answers, merged; CLEAN-019, CLEAN-020 merged. Nothing left in the lane.

Per stage

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
22:53	RES-001	impl	terra xhigh	16	10.7M	99%	27k	0.792	to-review
22:58	RES-001	review	sol xhigh	5	2.5M	95%	14k	0.637	merged
23:06	CORE-007	impl	terra xhigh	8	3.9M	98%	8k	0.299	to-review
23:09	CORE-007	review	sol xhigh	3	1.3M	94%	8k	0.374	merged
23:20	CLOCK-003	impl	terra xhigh	10	4.7M	98%	11k	0.384	to-review
23:23	CLOCK-003	review	sol xhigh	3	1.5M	93%	9k	0.437	merged
23:35	CI-005	impl	terra xhigh	12	6.1M	99%	17k	0.462	to-review
23:40	CI-005	review	sol xhigh	5	2.4M	95%	17k	0.657	merged
23:54	LEARN-001	impl	terra xhigh	14	10.6M	99%	25k	0.762	to-review
00:00	LEARN-001	review	sol xhigh	7	4.2M	95%	21k	1.016	merged
00:20	LEARN-002	impl	terra xhigh	19	14.0M	99%	35k	0.993	to-review
00:25	LEARN-002	review	sol xhigh	5	3.3M	95%	16k	0.772	reopened
00:33	LEARN-002	impl	terra xhigh	8	5.5M	99%	8k	0.386	to-review
00:36	LEARN-002	review	sol xhigh	3	1.6M	94%	10k	0.447	merged
00:54	LEARN-003	impl	terra xhigh	18	12.2M	99%	23k	0.826	to-review
00:58	LEARN-003	review	sol xhigh	4	1.9M	94%	11k	0.525	merged
01:16	LEARN-004	impl	terra xhigh	18	11.6M	99%	29k	0.834	to-review
01:26	LEARN-004	review	sol xhigh	10	4.7M	95%	23k	1.118	merged
01:46	LEARN-005	impl	terra xhigh	20	18.1M	99%	33k	1.192	to-review
01:50	LEARN-005	review	sol xhigh	5	2.8M	94%	15k	0.738	reopened
02:03	LEARN-005	impl	terra xhigh	12	7.5M	99%	13k	0.518	to-review
02:06	LEARN-005	review	sol xhigh	3	1.9M	95%	10k	0.491	merged
02:21	LEARN-006	impl	sol xhigh	15	3.5M	97%	17k	0.685	split, no code
02:28	CLEAN-016	impl	terra xhigh	8	1.9M	98%	5k	0.156	to-review
02:31	CLEAN-016	review	sol xhigh	3	1.1M	94%	8k	0.302	merged
02:40	CLEAN-017	impl	terra xhigh	9	4.7M	99%	10k	0.341	to-review
02:43	CLEAN-017	review	sol xhigh	3	1.3M	93%	8k	0.379	merged
03:05	LEARN-010	impl	sol xhigh	22	8.5M	98%	56k	1.739	to-review
03:09	LEARN-010	review	sol xhigh	4	2.7M	95%	14k	0.672	reopened
03:21	LEARN-010	impl	sol xhigh	12	4.7M	98%	20k	0.897	to-review
03:26	LEARN-010	review	sol xhigh	5	3.7M	94%	19k	0.983	reopened
03:46	LEARN-010	impl	sol xhigh	17	6.6M	98%	34k	1.316	to-review
03:50	LEARN-010	review	sol xhigh	4	3.1M	95%	13k	0.732	merged
04:13	LEARN-006	impl	sol xhigh	23	12.8M	98%	59k	2.301	to-review
04:17	LEARN-006	review	sol xhigh	4	3.1M	93%	14k	0.835	merged
04:42	LEARN-007	impl	terra xhigh	25	26.7M	99%	47k	1.747	to-review
04:58	LEARN-007	review	sol xhigh	16	7.9M	97%	31k	1.510	merged
05:20	LEARN-008	impl	terra xhigh	22	22.1M	99%	40k	1.471	parked (false positive)
05:30	CLEAN-018	impl	terra xhigh	10	3.2M	98%	9k	0.259	to-review
05:34	CLEAN-018	review	sol xhigh	3	1.3M	94%	9k	0.381	merged
05:47	LEARN-008	review	sol xhigh	11	8.5M	98%	22k	1.454	merged
06:19	LEARN-009	impl	terra xhigh	32	32.6M	99%	55k	2.234	to-review

When	Ticket	Stage	Model	Min	In	Cache	Out	Cost	Outcome
06:25	LEARN-009	review	sol xhigh	6	3.4M	95%	16k	0.843	merged
06:28	REV-011	impl	terra xhigh	3	1.2M	95%	5k	0.141	asked, blocked
06:43	REV-011	impl	terra xhigh	14	8.5M	99%	24k	0.648	to-review
06:47	REV-011	review	sol xhigh	4	2.4M	94%	13k	0.627	merged
07:00	CLEAN-019	impl	terra xhigh	12	4.4M	98%	12k	0.345	to-review
07:03	CLEAN-019	review	sol xhigh	3	1.5M	92%	8k	0.446	merged
07:17	CLEAN-020	impl	terra xhigh	14	10.6M	99%	25k	0.757	to-review
07:24	CLEAN-020	review	sol xhigh	6	2.4M	95%	17k	0.665	merged
				518	327.4M			39.53	



Wasted: the four reopened reviews and the stage 2 runs that answered them, $6.48 (LEARN-002 $1.16, LEARN-005 $1.26, LEARN-010 $4.07 across two rounds and three re-implementations, counting the extra review). LEARN-006's first attempt ($0.69) wrote no code but produced the split; REV-011's first attempt ($0.14) only asked. LEARN-008's park cost nothing: its work was kept and reviewed as handed off. The heaviest tickets by input were LEARN-009 (36.0M), LEARN-007 (34.6M) and LEARN-008 (30.6M): screens that reach Main, Home and the course module at once.

Comparison with previous runs

From `run-log.md`. `audit-fixes` (6 Nov, Windows): $6.07 for 6 ($1.01 each), no reopens. `review-explore` (5 Nov): $11.40 for 3 ($3.80 each), four reopens. This run: $39.53 for 20 ($1.98 each), 4 of 24 reviews reopened (17%). It is the longest P-Chess run so far (8 h47, three launches), and a feature lane, so it compares with `review-explore` rather than with the bounded fixes of `audit-fixes`.

Observations per model

- **gpt-6.1-terra xhigh, stage 2** (21 stages, 3–32 min): 15 tickets on its first attempt. Two reopens (LEARN-002, LEARN-005), both explicit criteria it missed. It took care of test-first strictly (LEARN-008: 14 commits, each pure). It asked well on REV-011: a real gap in the data, not a guess.
- **gpt-6.1-sol xhigh, stage 2 (hard)** (5 stages): LEARN-006 stopped before writing code and took the contract question to the proxy, which is what the ticket-flow asks. LEARN-010 needed three rounds: its first repair claimed start-up coverage without a test for it.
- **gpt-6.1-sol xhigh, review** (24 stages, 3–16 min): reopens were reproduced, with exact repair instructions. Two misses: LEARN-010's round 1 read malformed output only during a search (round 2 found the start-up path, already present), and LEARN-010 was approved with its own new Windows CI job never run. It filed five `CLEAN-*` tickets instead of widening scope, and the driver ran all five.

- **opus-5.5 xhigh, proxy**: six decisions (below). One call's answer started with a preamble line, and `proxy.ps1` rejected it, so REV-011's stage reported a proxy failure. The driver's second call gave the same value (3 moves) but put it in the data file instead of a code constant.

Driver and runtime issues

1. The test-file pattern in `driver.ps1` (`[._-](test|spec)\.[^/.]+$`) matched the production screen `ui/pass_test.gd`. LEARN-008 was parked as "incomplete handoff". The foreman checked the 14 commits and restored `to-review` (`9fa4730`).
2. LEARN-006's planned stop after the proxy's split was logged as `failed (exit 0), will retry` and spent attempt 1.
3. `proxy.ps1` needs `Decision:` on the first line; an answer with a preamble line counts as a failure even with a decision below it.
4. The main checkout's `.git/info/exclude` lacked the run-log, `*.uid` and `*.import` lines the Windows checkout has; the foreman added them to the new clone. After CI-005 and LEARN-010, the ignored `engine/tablebase` (a link to the hard disk) and `engine/tbprobe/` showed as untracked on `main`, whose `.gitignore` lacks them, and would have failed Preflight; added to the clone's exclude.
5. No Linux Lc0 and no display in the clone: six live Maia tests and the headless pixel checks were skipped in every gate; Xvfb was used for the clipboard and board captures.
6. No crash, timeout or API error. Shutdown 0.5–1.8 s per stage.

Restarts and cleanups

No crash restart. Two relaunches after foreman fixes: at 03:28, after the LEARN-010 reopen diagnosis (`c087ae0`), and at 05:35, after the LEARN-008 restore (`9fa4730`). The diagnosis classified the second reopen as an earlier review miss plus an incomplete repair, not a contract gap. Its consolidated review added three items: phase-aware validation, the exact `id name` line, and a non-draw with distance 0. Round 3 merged with no findings.

Noticed and left alone

- The proxy, not Bruno, approved the change to spec L5 (see Your turn).
- CLEAN-017's write and rename failure paths were checked by reading, with no injected failure (the ticket asked for no new tests).
- Several tickets wrote evidence to `/run/media/lage/NewVolume/Scratch/`, which is local to this machine; the tickets carry the commands.

Human debt and proxy decisions

No `Débito humano:` lines in this run's tickets. All six decisions were made by `opus-5.5 xhigh`; Codex sessions reached the proxy directly, and one (REV-011) was asked by the driver after the stage's own call failed to parse.

- RES-001 (22:39): `Proxy decided (opus-5.5 xhigh): proceed on Linux and do not wait for a Windows run.` Criterion 2 measured with the Linux build of Stockfish 19; Windows syntax pending Bruno; lc0 built natively for criterion 5.
- LEARN-002 (00:07): `Proxy decided (opus-5.5 xhigh): Keep all four parts on the unit page and the course-map card; reword criterion 5 so it bans placeholder units and stages, not the four fixed parts.`
- LEARN-003 (00:43): `Proxy decided (opus-5.5 xhigh): LEARN-003 adds no book or chapter entries.` LEARN-004 supplies them.

- LEARN-004 (00:59): `Proxy decided (opus-5.5 xhigh): Name the stage ``Endgame foundations'' and go ahead with the draft content.`
- LEARN-006 (02:17): `Proxy decided (opus-5.5 xhigh): split (B).` The native probe becomes LEARN-010; LEARN-006 waits for it.
- REV-011 (06:28, asked by the driver): `Proxy decided (opus-5.5 xhigh)`: add `"default_line_length":3` to the unit file and use it for every "From your game" position.

Receipts: `/tmp/sweatshop-proxy/` on this machine, named in each ticket.

Acceptance

Spec scenarios, on the integrated session (`66abb54`, gate 677 passed, 0 failed; real tablebase tests ran): scenario 4 (review check at 7 days) passed in `test_continue_offers_review_checks_at_seven_and_thirty_days` and two screen tests. Scenarios 1, 2, 3 and 5 are Bruno's hand checks (Your turn); their automated parts pass. Not examined: anything on Windows, a real Maia reply, and a person at the screen. Recorded in `.scratch/learning/spec.md` and PR #44.

Reopen attribution

Labelled by the foreman (opus-5.5); every round's reviewer was `gpt-6.1-sol xhigh`, so none is self-judged. Rows appended to `.scratch/reopen-attribution.md`.

Ticket (round)	Finding	Label	Why
LEARN-002 (1)	a pending review check hides a failed one; "needs review" vanishes	S2-explícito/código	criterion 2; no test had the case
LEARN-005 (1)	move numbers wrong when Black starts	S2-explícito/código	criterion 5
LEARN-005 (1)	Enter wipes an unfinished line while Clear has focus	S2-explícito/código	criterion 7
LEARN-010 (1)	malformed `bestmove` tokens and unknown lines during a search succeed	S2-explícito/código	criterion 4
LEARN-010 (2)	invalid lines before the handshake barriers or while idle are dropped	S2-explícito/código	criterion 4; present since the first handoff: review miss (drip-feed)
LEARN-004 (0)	the shape test did not require both goals in each pool	S2-explícito/teste	added by the review (`fdf31ff`)
LEARN-010 (–)	Windows smoke probe fails in CI	unclassified	post-approval, `session-review`

Totals: S2-explícito/código 5, S2-explícito/teste 1 (fixed in review), unclassified 1 (post-approval). No S1 or S3. Review miss: LEARN-010 round 2. No review-induced churn.

Scoreboard

`scoreboard.ps1` over this machine's main checkouts (ClassyCall, EasyMath, P-Chess, PhysiSyst, SIGAA-ME, SynchroNice, slip), with P-Chess read from the lane clone, whose log is identical to the main checkout's after the copy-back and which alone holds this run's attribution and interventions. Moved since the `review-explore` report (the last one from this machine; `audit-fixes` ran on Windows): pair `gpt-6.1-terra xhigh` → `gpt-6.1-sol xhigh` 101 → 154 reviews, 29% → 24% reopened (S2 ≥23%); pair `gpt-6.1-sol xhigh` → `gpt-6.1-sol xhigh` 22 → 33 reviews, 23% → 27% (LEARN-010's two reopens); proxy-decision (opus-5.5 xhigh) 47 → 64; automatic-recovery 130 → 133.

Implementer	Stage runs	To review	Failed	Asked	Median min to review	Cost
gpt-6-astra medium	57	47	1	9	10m	$162.72 (57/57 priced)
gpt-6.1-sol-fast high	18	16	0	2	10m	$21.98 (15/18 priced)
gpt-6.1-sol-fast max	14	12	0	2	12m	$31.19 (14/14 priced)
opus-5.5	42	35	4	3	8m	$15.84 (8/42 priced)
gpt-6-luna max	89	61	10	18	17m	$4.91 (85/89 priced)
sonnet-5.5 high	18	17	1	0	7m	$31.92 (18/18 priced)
gpt-6.1-sol high	311	275	12	24	23m	$839.52 (309/311 priced)
gpt-6-astra high	1	0	0	1	-	$3.92 (1/1 priced)
opus-5.5 xhigh	24	16	8	0	14m	$100.17 (22/24 priced)
gpt-6.1-sol max	41	29	9	3	24m	$84.42 (33/41 priced)
haiku-5.5 high	26	6	15	5	21m	$72.08 (24/26 priced)
haiku-5.5 xhigh	4	0	3	1	-	$12.29 (4/4 priced)
opus-5.5 high	37	21	15	1	12m	$73.54 (29/37 priced)
gpt-6.1-sol xhigh	44	33	1	0	19m	$99.35 (44/44 priced)
gpt-6.1-terra high	59	30	27	2	9m	$9.45 (34/59 priced)
gpt-6.1-terra xhigh	185	155	19	11	15m	$139.31 (185/185 priced)
sonnet	121	75	31	15	7m	$10.53 (4/121 priced)
sonnet-5.5 xhigh	9	8	1	0	25m	$35.54 (9/9 priced)
fable	5	5	0	0	3m	?
gpt-6-luna	23	11	10	2	26m	?
gpt-6-sol	6	5	1	0	17m	?
gpt-5.6-sol	5	4	1	0	17m	?
gpt-6-astra	18	11	6	1	8m	?
sonnet-5 xhigh	11	6	2	3	8m	$18.09 (11/11 priced)
gpt-6-astra low	2	2	0	0	30m	$19.03 (2/2 priced)

Implementer	Reviewer	Reviews	Reopened	Reopen rate	Reopened for S2	S2 rate	Unclassified	Fixed in review for S2
gpt-6-astra medium	gpt-6.1-sol max	37	7	19%	6	16%	0	1
gpt-6.1-sol-fast high	gpt-6.1-sol-fast max	16	2	12%	2	12%	0	0
gpt-6.1-sol-fast max	gpt-6.1-sol-fast max	12	6	50%	5	42%	0	0
opus-5.5	opus-5.5	4	0	0%	0	0%	0	0
gpt-6-luna max	opus-5.5 high	18	1	6%	1	6%	0	0
gpt-6-luna max	gpt-6-sol max	43	23	53%	22	51%	0	0
?	gpt-6-sol max	1	0	0%	0	0%	0	0
sonnet-5.5 high	gpt-6-sol max	2	1	50%	1	50%	0	0
gpt-6.1-sol high	gpt-6.1-sol max	94	21	22%	20	21%	0	4
opus-5.5 xhigh	gpt-6.1-sol max	6	3	50%	3	50%	0	0
?	gpt-6.1-sol max	4	0	0%	0	0%	0	1
gpt-6.1-sol max	gpt-6.1-sol max	19	5	26%	5	26%	0	0
haiku-5.5 high	gpt-6.1-sol max	6	2	33%	2	33%	0	0
gpt-6.1-sol high	gpt-6.1-sol xhigh	176	33	19%	32	18%	0	20
gpt-6.1-sol max	gpt-6.1-sol xhigh	9	2	22%	2	22%	0	0
opus-5.5 xhigh	opus-5.5 xhigh	9	0	0%	0	0%	0	0
opus-5.5 high	opus-5.5 xhigh	13	2	15%	2	15%	0	5
gpt-6.1-sol xhigh	gpt-6.1-sol xhigh	33	9	27%	9	27%	0	1
opus-5.5 high	gpt-6.1-sol xhigh	1	1	100%	1	100%	0	0
gpt-6.1-terra high	gpt-6.1-sol xhigh	30	7	23%	7	23%	0	0
?	gpt-6.1-sol xhigh	8	2	25%	1	12%	0	0
gpt-6.1-terra xhigh	gpt-6.1-sol xhigh	154	37	24%	36	≥23%	1	15
?	opus-5.5 xhigh	4	1	25%	1	25%	0	0
sonnet	opus	68	20	29%	1	≥1%	19	0
?	opus	4	0	0%	0	0%	0	0
opus-5.5	fable-5.1	30	1	3%	0	≥0%	1	0
sonnet-5.5 high	gpt-6.1-sol max	12	3	25%	3	25%	0	0
sonnet-5.5 xhigh	gpt-6.1-sol max	7	3	43%	3	43%	0	0
opus-5.5 high	fable-5.1 high	1	0	0%	0	0%	0	0
sonnet-5.5 high	opus-5.5 xhigh	3	0	0%	0	0%	0	3
sonnet-5.5 xhigh	opus-5.5 xhigh	1	0	0%	0	0%	0	0

Implementer	Reviewer	Reviews	Reopened	Reopen rate	Reopened for S2	S2 rate	Unclassified	Fixed in review for S2
gpt-6-astra medium	gpt-6-astra xhigh	10	1	10%	1	10%	0	0
sonnet	fable	1	0	0%	0	0%	0	0
fable	fable	4	1	25%	0	≥0%	1	0
gpt-6-luna	gpt-6-sol	11	7	64%	0	≥0%	7	0
gpt-6-sol	gpt-5.6-sol	5	4	80%	0	≥0%	4	0
gpt-5.6-sol	gpt-6-astra	4	3	75%	0	≥0%	3	0
gpt-6-astra	gpt-5.6-sol	11	5	45%	0	≥0%	5	0
sonnet-5 xhigh	opus-5.5 xhigh	5	0	0%	0	0%	0	0
gpt-6-astra low	gpt-6.1-sol max	2	1	50%	1	50%	0	0

Implementer → Reviewer	Rounds	S2 explicit code	S2 explicit test	S2 implicit	S1	S3 noise
gpt-6-astra low → gpt-6.1-sol max	1	2	0	1	0	0
gpt-6-astra medium → gpt-6-astra xhigh	1	1	0	0	0	0
gpt-6-astra medium → gpt-6.1-sol max	8	11	1	4	1	0
gpt-6-luna max → gpt-6-sol max	23	31	3	23	12	15
gpt-6-luna max → opus-5.5 high	1	1	1	0	1	0
gpt-6.1-sol high → gpt-6.1-sol max	26	16	4	23	1	0
gpt-6.1-sol high → gpt-6.1-sol xhigh	54	54	4	23	3	0
gpt-6.1-sol max → gpt-6.1-sol max	5	7	0	5	0	0
gpt-6.1-sol max → gpt-6.1-sol xhigh	2	0	0	3	0	0
gpt-6.1-sol xhigh → gpt-6.1-sol xhigh	9	20	0	5	0	0
gpt-6.1-sol-fast high → gpt-6.1-sol-fast max	2	2	0	0	0	0
gpt-6.1-sol-fast max → gpt-6.1-sol-fast max	6	2	0	3	1	0
gpt-6.1-terra high → gpt-6.1-sol xhigh	9	10	1	3	1	0
gpt-6.1-terra xhigh → gpt-6.1-sol xhigh	51	72	4	14	5	0
haiku-5.5 high → gpt-6.1-sol max	3	3	1	5	1	0
opus-5.5 high → gpt-6.1-sol xhigh	1	2	0	0	0	0
opus-5.5 high → opus-5.5 xhigh	8	7	0	3	0	0
opus-5.5 xhigh → gpt-6.1-sol max	3	3	0	2	1	0
sonnet → opus	1	1	1	0	0	0
sonnet-5.5 high → gpt-6-sol max	1	3	0	2	0	0
sonnet-5.5 high → gpt-6.1-sol max	3	5	0	2	3	0
sonnet-5.5 high → opus-5.5 xhigh	3	1	2	0	0	0
sonnet-5.5 xhigh → gpt-6.1-sol max	3	7	0	3	2	0

Evidence coverage: legacy flags and absent records are unknown, not zero defects or zero interventions.

Discovery	Recorded findings	Earlier review miss (yes / known)	Review-induced churn (yes / known)
release-audit	7	2 / 2	0 / 2
session-review	37	30 / 32	0 / 32
ticket-review	439	10 / 300	1 / 302

Runtime completion to process exit: 1746.941s total over 1338 measured stages; 773 unmeasured.

Intervention	Recorded events
automatic-recovery	133
human-rescue	33
proxy-decision (gpt-6-astra high, fallback)	58
proxy-decision (opus-5.5 xhigh)	64
proxy-decision (opus)	42
proxy-decision (supervisor)	12

Intervention	Recorded events
proxy-decision (unknown)	37
routine-approval	20

Under the same reviewer, Terra xhigh (154 reviews) now reopens less than Sol xhigh implementing (33 reviews, hard tickets only), so the comparison is not like for like. Lineup: keep the default; the rows above are the evidence.